

# July 3 Action Plan

Empirical variances & Monte-Carlo standard errors; and normalizing the ability distribution  $\ell$  to uniform

July 8, 2026

This plan has three coupled workstreams. **Part I** fixes how the empirical moment variances are computed and used (the weight matrix and the Monte-Carlo standard errors). **Part II** replaces the estimated Beta ability distribution  $\ell$  with a uniform normalization—and documents why that change, on its own, is necessary but not sufficient. **Part III** specifies the documentation that must follow: the edits to the *Model Notes* and the *Data and Moments* notes, and a new companion note, *Transition and MCMC*, written in the same style.

## 1 Part I — Empirical variances, weighting, and standard errors

### Decisions locked in

- **$\hat{\Sigma}$  estimator:** a *design-based replicate-weight bootstrap* of the data moments, in *level* units. Highest standard and most defensible in print (the search-SMM literature, incl. LMR, bootstraps; official survey replicate weights are the Census/BLS-sanctioned method).
- **Weighting:** *diagonal only*,  $W = \text{diag}(1/\hat{\sigma}_k^2)$ ; never the full matrix. Only the diagonal of  $\hat{\Sigma}$  is used as the weight. The bootstrap still yields the full  $\hat{\Sigma}$  for free—keep it in its branch, it is what a correct sandwich SE consumes.
- **JOLTS/J2J moments:** use the *published* standard errors (sources below); verify they match the downloaded files.

### Diagnosis of the current pipeline

| Issue                                              | Where                                                                         | Consequence                                                                                            |
|----------------------------------------------------|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Period (time-series) variance, not cross-sectional | <code>sigma.jl</code> ( $\Psi$ rows = periods, $\hat{\Sigma} = \Psi'\Psi/T$ ) | Conflates business cycle with sampling noise; contradicts “steady state $\Rightarrow$ cross-sectional” |
| No HAC                                             | $\hat{\Sigma} = \Psi'\Psi/T$ treats quarters i.i.d.                           | Understates even the time-series variance (moments autocorrelated)                                     |
| Relative units vs level deviations                 | <code>sigma.jl</code> <code>Psi ./= scales</code> ; loss uses raw deviations  | Injects a spurious $\bar{m}_k^2$ per moment (the “Q1” bug)                                             |
| Per-period $\hat{\sigma}$ in the quasi-posterior   | <code>MCMC_main.jl</code> ( $-\frac{1}{2}g'Wg$ )                              | Chain SD $\neq$ sampling SE; $\sim\sqrt{T}$ too diffuse                                                |

**Contrast (LMR, RED).** `bsample`, `cluster(id)` (resample persons), recompute all moments, SD across replications  $\rightarrow$  `se_ed1.raw`; criterion  $-\frac{1}{2}\sum_k ((\text{model}_k - \text{data}_k)/\text{SD}_k)^2$  with *level* deviation  $\div$  *level* SE; MCMC (Chernozhukov–Hong) reads SEs off the chain. Clean because SD is the sampling SE of the estimate and units match.

## Target design

$\hat{\Sigma}$  = the sampling variance–covariance of the data moment vector  $\hat{m}$ , in *level* units, per estimation window, via the survey-design replicate-weight bootstrap:

$$\hat{\Sigma} = \frac{c}{R} \sum_{r=1}^R (\hat{m}_r - \hat{m})(\hat{m}_r - \hat{m})' \quad (\text{successive-difference replication; confirm } c, R \text{ vs. IPUMS docs}).$$

This automatically handles ratios (`ur`), quantiles (`p25/p50`), two-sample overlaps and duration shares (`ltu`), and captures within-person serial dependence (replicate weights are person-level). One free advantage: the model moments are *analytic* steady-state objects, so there is *zero* simulation noise—no  $(1 + 1/S)$  inflation (unlike LMR).

## Action items

### Phase 0 — immediate correctness (small, do first).

- ☐ **A0.1** Fix the loss units: divide level deviations by a level SD. One-liner in `compute_loss_matrix` (divide `dev_vec` by `moment_scales`) *or* `load_weight_matrix` (multiply  $\hat{\Sigma}$  by `scales·scales'` before use).
- ☐ **A0.2** Make the MCMC weight equal the estimation weight (else `post_mean`  $\neq$  `point_estimate`).

### Phase 1 — build the bootstrap $\hat{\Sigma}$ (core).

- ☐ **A1.0** Refactor moment computation into a pure function callable on any (replicate-weighted) dataset.
- ☐ **A1.1** CPS Basic replicate weights (`REPWTP`); recompute stocks (`ur·U/S`, shares, `ltu_share_S`) and flows (`jfr`, `sep`)—resample by person (`CPSIDP`), not person-month.
- ☐ **A1.2** CPS ASEC replicate weights; recompute wage moments, `wage_premium`, overlaps.
- ☐ **A1.3** JOLTS `theta·U/S`: published SE (see provenance).  $\theta = V/U$ , so combine the JOLTS job-openings SE with the CPS  $U$  variance; JOLTS dominates.
- ☐ **A1.4** J2J `ee_rate_S`: published SE `sEEHireR` (rate) or `sEEHire` (count).
- ☐ **A1.5** Assemble  $\hat{\Sigma}$  in *level* units; drop the `Psi ./= scales` relativization; keep `moment_scales` for reporting only.
- ☐ **A1.6** Validate: analytic-IF spot-check on one mean and one rate; bootstrap SEs should be *much smaller* than the old period dispersion.

### Phase 2 — wire into estimation & inference.

- ☐ **A2.1** `load_weight_matrix` reads the level  $\hat{\Sigma}$ ;  $W = \text{diag}(1/\hat{\sigma}^2)$ .
- ☐ **A2.2** Loss = level deviation  $\div$  level SD (consistent with A0.1).
- ☐ **A2.3** Report SEs via the sandwich  $\widehat{\text{Var}}(\hat{\theta}) = (D'WD)^{-1}D'W\hat{S}WD(D'WD)^{-1}$  with the full bootstrap  $\hat{\Sigma}$  as  $\hat{S}$  (valid for any  $W$ , incl. diagonal). The DE-MC chain SD is a cross-check only (valid iff  $W = \hat{\Sigma}^{-1}$ ).

**Phase 3 — referee-proofing.** Document the replicate-weight formula ( $c, R$ ); note the CPS Basic $\leftrightarrow$ ASEC overlap (their cross-covariance is not zero, but the diagonal weight never uses it); optionally keep the old period variance as an appendix “target uncertainty” robustness column.

## Published-SE provenance (verify against your downloads)

- **JOLTS Median Standard Errors** (Balanced-Half-Sample; *median* of monthly NSA SEs over the prior 5 years—*not* window-specific; use the job-openings row for  $V$ ): [https://www.bls.gov/jlt/jolts\\_median\\_standard\\_errors.htm](https://www.bls.gov/jlt/jolts_median_standard_errors.htm); national table <https://www.bls.gov/jlt/jlt-mse2025.xlsx>; method <https://www.bls.gov/jlt/jltreliability.htm>.
- **J2J** carries an SE for every estimate as an **s**-prefixed twin (**sEEHire**, **sEEHireR**); cell-specific, so window-specific for free. Re-pull including the **sE\*** columns from the same **education=="E4"** cell. Schema [https://lehd.ces.census.gov/data/schema/V4.2-rc1/lehd\\_j2jexplorer\\_schema.html](https://lehd.ces.census.gov/data/schema/V4.2-rc1/lehd_j2jexplorer_schema.html); data <https://lehd.ces.census.gov/data/>; method [https://lehd.ces.census.gov/doc/j2j\\_101.pdf](https://lehd.ces.census.gov/doc/j2j_101.pdf).

## Resolved decisions

(1) JOLTS/J2J  $\rightarrow$  published SEs. (2) Weighting  $\rightarrow$  diagonal only. (3)  $R$  is fixed by the survey (use all replicate weights); compute a *separate*  $\hat{\Sigma}$  per window. (4) Cross-source covariance is not zero (esp. CPS–ASEC), but is moot under a diagonal weight; keep full  $\hat{\Sigma}$  in its branch for the sandwich.

## 2 Part II — Normalize $\ell$ to uniform

### Why: the estimated $\ell$ is degenerate

The point estimate has  $\ell = \text{Beta}(a_\ell=2.59, b_\ell=0.32)$ . Since  $b_\ell < 1$ ,  $\ell$  has an integrable *singularity* at  $x = 1$ : in the port,  $\ell(1.0) \approx 649$ , and 34% of the population sits at  $x \geq 0.99$ . The training cutoff is  $\bar{x} \approx 0.99$ : essentially only the top ability sliver trains. This is not a credible ability distribution—it is an artifact.

**Mechanism (confirmed in the port).** The cutoff is pinned near 1 by the *specification*, not by  $\ell$ : (i) the training cost  $c(x) = (1 - x)e^{\hat{c}-x}$  (with  $\hat{c} = 11.1$ ) is  $\sim 2,700$  at  $x = 0.9$  but  $\sim 24$  at  $x = 0.999$ —affordable only at the very top; and (ii) skilled productivity  $P_S(x) = \gamma_{PS}x^{\gamma_{PS}-1}$  exceeds  $P_U$  only above  $x \approx 0.956$  ( $\gamma_{PS} = 9.72$  is steep). Given a cutoff at 0.99, the only way to a 37% skilled share is for  $\ell$  to pile a third of the mass into  $[0.99, 1]$ , i.e.  $b_\ell < 1$ .

**Identification.** In sorting/matching models the type distribution and the productivity map are *not separately identified*:  $x$  enters only through  $P_S(x) \cdot x$ , so a monotone relabeling of  $x$  can be absorbed by reshaping  $\ell$  and  $P_S$  together. The literature normalizes types to *uniform ranks* and puts all content in the production function (Hagedorn–Law–Manovskii, *Econometrica* 2017; Lise–Meghir–Robin, *RED*). Estimating  $a_\ell, b_\ell$  and  $\gamma_{PS}$  freely is over-parameterized; the objective resolves the indeterminacy at a boundary ( $b_\ell \rightarrow 0$ ). **Hence: set  $\ell \equiv \text{Uniform}[0, 1]$  (=  $\text{Beta}(1, 1)$ ) and drop  $a_\ell, b_\ell$  from the free block.**

## Evidence from the port (why uniform alone is not enough)

Setting  $\ell = \text{Uniform}$  and searching  $c, \gamma_{PS}, P_U$  (grid  $N=60$ ):

| config ( $\ell$ = Unif unless noted)            | $\bar{x}$   | skl. share  | $\bar{w}_U$ | $Q$  | note                              |
|-------------------------------------------------|-------------|-------------|-------------|------|-----------------------------------|
| baseline $\ell$ = Beta(2.59, 0.32)              | 0.988       | 0.335       | 6.09        | 1.82 | degenerate $\ell$ (spike)         |
| uniform, else current                           | 0.985       | 0.016       | 5.79        | 31.8 | skilled sector nearly gone        |
| uniform, $P_U=4$                                | 0.781       | 0.072       | 5.46        | 140  | both sectors alive, share too low |
| uniform, $P_U=6$                                | 0.960       | 0.031       | 5.72        | 39.1 | cutoff back near the top          |
| uniform, $P_U=1.2, \gamma_{PS}=2$               | 0.000       | 1.000       | 0.00        | huge | everyone trains; unskilled dies   |
| uniform, $P_U \downarrow$ (interior $\bar{x}$ ) | $\sim 0.38$ | $\sim 0.60$ | 0.00        | huge | unskilled sector collapses        |

**Key finding.** Under uniform  $\ell$  there is *no* parameter configuration that delivers, simultaneously, (a) a healthy unskilled sector, (b) an interior cutoff, and (c) a skilled share near 0.37. A healthy unskilled sector needs  $P_U$  moderate/high, which (because  $P_S(x) = \gamma_{PS} x^{\gamma_{PS}-1}$  only gets large near  $x = 1$ ) forces the productivity crossover to the top and the share to 0.03–0.07; pushing the cutoff down to  $\bar{x} \approx 0.63$  (needed for a 37% share) requires  $P_U$  so low the unskilled labor market dies ( $\bar{w}_U = 0$ ). The Beta spike was papering over this. **So normalizing  $\ell$  is necessary but not sufficient: it exposes a structural inadequacy in the skilled production function.**

### The structural change uniform $\ell$ forces

Skilled output is currently  $A P_S(x) x p = A \gamma_{PS} x^{\gamma_{PS}} p$ , so skilled productivity is negligible except near  $x = 1$ . To make training valuable over a *broad* range of abilities (so an interior cutoff can coexist with a live unskilled sector and a 37% share), add a *return to skill that is not the steep  $x^{\gamma_{PS}}$  gradient*, e.g.:

- a proportional training premium: skilled output =  $A P_U x p (1 + \pi)$ ,  $\pi > 0$ ; or
- an additive/level shifter:  $P_S(x) = P_{S0} + \gamma x$  with  $P_{S0} > P_U$ ; or
- a less convex gradient (small  $\gamma_{PS}$ ) *together with* a level term so  $P_S(x) > P_U$  over an interior range.

### Action items (Part II)

- ☐ **B1** Set  $\ell \equiv$  Uniform: in `default_free_params`, move  $a_\ell, b_\ell$  to the fixed block at 1.0 (like  $r, \nu, \phi$ ). *No grid change needed*—`build_type_grid(N, 1, 1)` reduces to Gauss–Legendre automatically.
- ☐ **B2** Add a return-to-skill term to the skilled production function (one of the three forms above) with its own estimated parameter; re-derive the skilled surplus/wage and the training-value  $T(x)$  accordingly (notes §5, §6.3, §8).
- ☐ **B3** (Optional) Replace the training cost  $c(x) = (1 - x)e^{c-x}$  with a form that does not vanish at  $x = 1$  (e.g.  $e^c(1 - x)$  or  $e^{c-\kappa x}$ ), so the cutoff is not mechanically pinned to the ceiling.
- ☐ **B4** Re-center bounds/start values for  $c, \gamma_{PS}, P_U, b_T, A$  (they all move together once  $\ell$  is uniform) and re-estimate.
- ☐ **B5** Report, for each variant:  $\bar{x}$ , skilled share, and the wage-*variance* fit. The dispersion moments (`emp_var_S`  $\approx 46\%$  of  $Q$ , `emp_var_U`  $\approx 37\%$ , `1tu_share_S`  $\approx 24\%$ ) are the headline shortfall and are *not* fixed by any of the above—they need more heterogeneity (match quality  $p$ , OJS dispersion, or a firm dimension).

## Does uniform $\ell$ give “better” estimates?

- **Credibility / identification: yes.**  $\ell$  becomes a principled normalization (ranks) rather than a fitted artifact; the  $b_\ell < 1$  degeneracy is gone; it is the field convention and is one line in the write-up.
- **Fit ( $Q$ ): no, and it will get worse first**—uniform  $\ell$  with today’s production function cannot manufacture the skilled sector (best non-degenerate points: share 0.03–0.07,  $Q \approx 40$ –140 vs. 1.82). The value is that it *reveals* the real modeling problem (skill return + wage dispersion), which is where the paper’s contribution should be.

## 3 Part III — Documentation: notes to edit and the note to create

The Part I and Part II changes make three documentation deliverables necessary, all in the shared LaTeX preamble/style of the existing *Model Notes* and *Data and Moments* notes. This part is a specification of *what* changes and *where*; the notes themselves are edited/written separately once the specification (Part II) and estimator (Part I) are settled.

### III.a — Edit *Model Notes*

- **Type distribution  $\rightarrow$  uniform (§1.1 “Aggregate scale...”, §2.3 deep-parameter table, §3.1 “Workers”).** Replace  $x \sim \text{Beta}(a_\ell, b_\ell)$  with  $x \sim \text{Uniform}[0, 1]$  (types are ranks). Remove  $a_\ell, b_\ell$  from the deep-parameter table. Add a short identification paragraph: in sorting/matching models the type distribution and the productivity map are *not separately identified*, so types are normalized to uniform ranks (Hagedorn–Law–Manovskii, *Ecta* 2017; LMR).
- **Return-to-skill in production (§5 “Production technologies”; §8.6 surplus, §8.7 wages; §6.3 training value  $T(x)$ ).** Add the skill-return term chosen in Part II (proportional premium  $\pi$ , i.e.  $\pi_S = A P_U x p(1 + \pi)$ , or an additive shifter  $P_{S0}$ ) to the skilled production function; re-derive the skilled surplus, the wage schedules, and  $T(x)$ . Without it a uniform  $\ell$  cannot sustain a skilled sector (Part II evidence).
- **Aggregate scale on a log scale (§1.1, §5).** The code now uses  $\exp(A)$  as the effective dollar scale ( $A$  searched on a log scale). Update the “ $b_j = \tilde{b}_j A$ ,  $k_j = \tilde{k}_j A$ ” convention to  $\exp(A)$ , and state explicitly that the training cost  $c(x)$  is *not* scaled by  $\exp(A)$  in the current code.
- **Training cost (§4), optional.** If reparametrized (Part II B3) to a form that does not vanish at  $x = 1$  (e.g.  $e^c(1 - x)$  or  $e^{c-\kappa x}$ ), replace  $c(x) = (1 - x)e^{c-x}$  and note the cutoff is no longer pinned to the ceiling.
- **Training-cutoff property (§13.1, Prop. 1).** Update the intuition for the uniform- $\ell$ , return-to-skill specification: the cutoff is now interior and set by the cost/return balance, not by an  $\ell$  spike.
- **Worker-type grid (numerical remark).** Note the type grid is Gauss–Jacobi tuned to  $\ell$ ; with  $\ell$  uniform it reduces to Gauss–Legendre.

### III.b — Edit *Data and Moments*

- **Variance construction (Part V, §“Influence functions”).** Replace the analytic-influence-function / per-period-deviation  $\hat{\Sigma}$  with the design-based replicate-weight bootstrap of Part I. The current text ( $\hat{\Sigma}_w = \frac{1}{N_w} \sum_i \psi \psi'$  under the “per-period-deviation convention  $\psi = \text{value}_{\text{period}} - \hat{m}$ ”) describes a *period/time-series* dispersion, not the cross-sectional sampling variance a steady-state model implies — this is the core correction.
- **Weight matrix (Part V, §“SMM weight matrix”).** Change from “identity headline, diagonal/full robustness” to *diagonal only*  $W = \text{diag}(1/\hat{\sigma}^2)$  in level units; state that parameter SEs come from the sandwich with the full bootstrap  $\hat{\Sigma}$  as the meat, and that JOLTS/J2J use *published* SEs (Part I provenance).
- **Identification of  $a_\ell, b_\ell$  (§“Identification”; §“Deep vs. regime-specific”).** Remove —  $\ell$  is now a normalization. State that `skilled_share` and `wage_premium` now identify the training-cost coefficient  $c$  and the return-to-skill parameter, not the ability distribution.
- **$\kappa_w$  training-share adjustment (§“Labour-market stocks”, §“Influence functions”).** Keep the NSC level anchor, but recompute  $\kappa_w$  *within each bootstrap replicate* so its uncertainty flows into  $\hat{\Sigma}$ ; this supersedes the A1–A3 plug-in variance argument and the Stage-8 row/column  $\kappa_w$ -scaling of  $\hat{\Sigma}$ .
- **File layout / pipeline.** `sigma_{window}.csv` now stores the bootstrap  $\hat{\Sigma}$  in *level* units (drop the relative `Psi ./= scales` step; keep `moment_scales` for reporting only). Note `sigma.jl` is replaced by a bootstrap routine.

### III.c — Create *Transition and MCMC* (new note)

A new companion note, same preamble/style, in two parts.

**Part A — Transition dynamics** (consolidate and expand *Model Notes* §12):

- the transition-equilibrium object; time-dependent HJBs given  $(\theta_U(t), \theta_S(t))$ ;
- the discrete cross-market (DC) policy  $d(x, t)$  activating along the path (it is  $\equiv 0$  in steady state, so this is where it matters);
- laws of motion for  $u_U, t, u_S, e_S$ ; aggregate mass dynamics and the cross-market drain; the switch initial condition at  $T$ ;
- free-entry tightness paths with the augmented seeker pools;
- the forward–backward numerical algorithm (backward HJB pass, forward density pass, tightness update, convergence);
- adjustment speed and the FC-vs-COVID reading (movement along vs. outward Beveridge shift).

**Part B — Estimation and inference (MCMC)** (new material, drawing on Part I of this plan):

- the SMM objective  $Q_w(\theta) = g'Wg$  and the multi-regime strategy (deep vs. regime-specific blocks); the optimizers (SA / DE / Nelder–Mead) and run diagnostics (per-parameter corner tags with lower/upper bound, the no-improvement early stop);
- the DE-MC / Chernozhukov–Hong Laplace-type quasi-posterior  $\pi(\theta) \propto \exp(-\frac{1}{2} g'Wg)$ ; the scaling, and why the variance *object* (bootstrap  $\hat{\Sigma}$ , level units, with  $N$  inside the SE) governs the posterior concentration;

- parameter standard errors: the chain covariance versus the sandwich formula (Part I, A2.3); that the chain SD is valid only at  $W = \hat{\Sigma}^{-1}$ , so under the diagonal weight the *sandwich* (with the full bootstrap  $\hat{S}$ ) is the reported SE;
- a normalization remark tying back to Part II:  $\ell$  uniform, and the  $\ell$ - $P_S$  non-identification that motivates it.

## 4 Suggested sequencing

1. Part I, Phase 0 (unit fix, matched weights)—makes current runs correct.
2. Part II B1–B2 (uniform  $\ell$  + a skill-return term); re-estimate (B4).
3. Part I, Phases 1–3 (bootstrap  $\hat{\Sigma}$  + sandwich SEs) on the new specification.
4. Report the wage-dispersion fit (B5) and decide the next structural step.
5. Part III documentation pass *last*, once the specification and estimator are settled: edit the *Model Notes* (III.a) and *Data and Moments* (III.b), and write the *Transition and MCMC* note (III.c).

**Sources.** Hagedorn, Law & Manovskii, “Identifying Equilibrium Models of Labor Market Sorting,” *Econometrica* 2017, [https://www.sas.upenn.edu/~manovski/papers/Identifying\\_Sorting.pdf](https://www.sas.upenn.edu/~manovski/papers/Identifying_Sorting.pdf); Lise, Meghir & Robin, “Matching, sorting and wages,” *RED*, <https://www.sciencedirect.com/science/article/pii/S109420251500071X>.